Open World aware Universal Negative Sampling for Knowledge Graph Completion

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MINDS Lab

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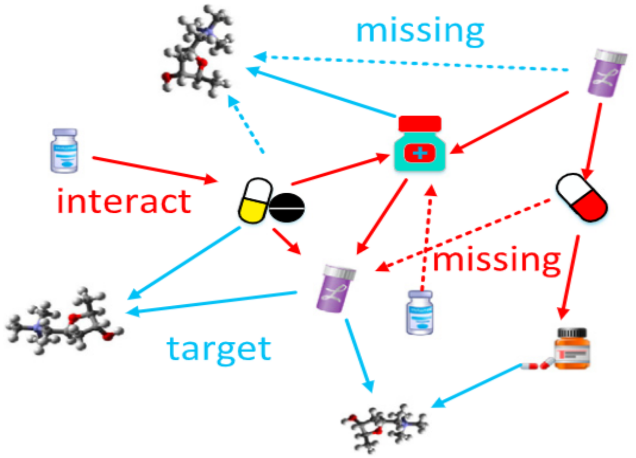
KNOWLEDGE GRAPH COMPLETION

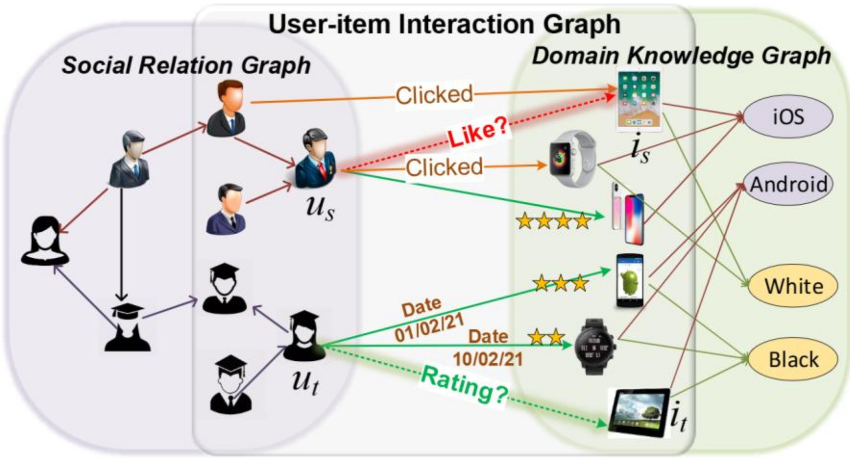
■ Knowledge Graph: Multi-relationship Between Entities

☐ **Edge : Triple** (head entity, relation, tail entity)

<Drug Discovery>



<Recommender System>



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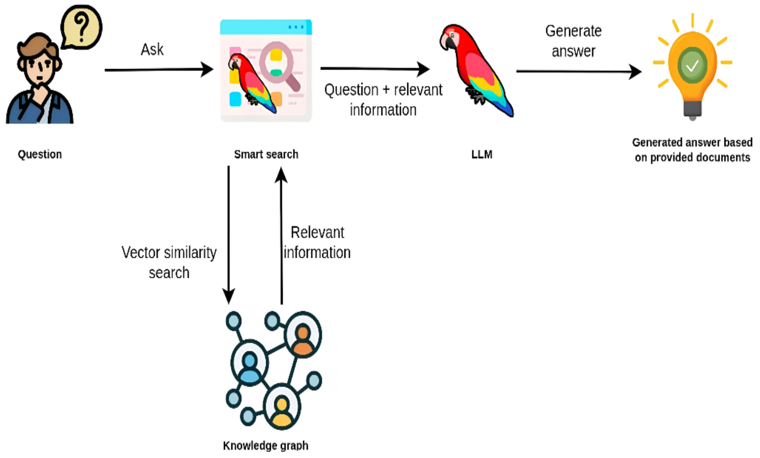
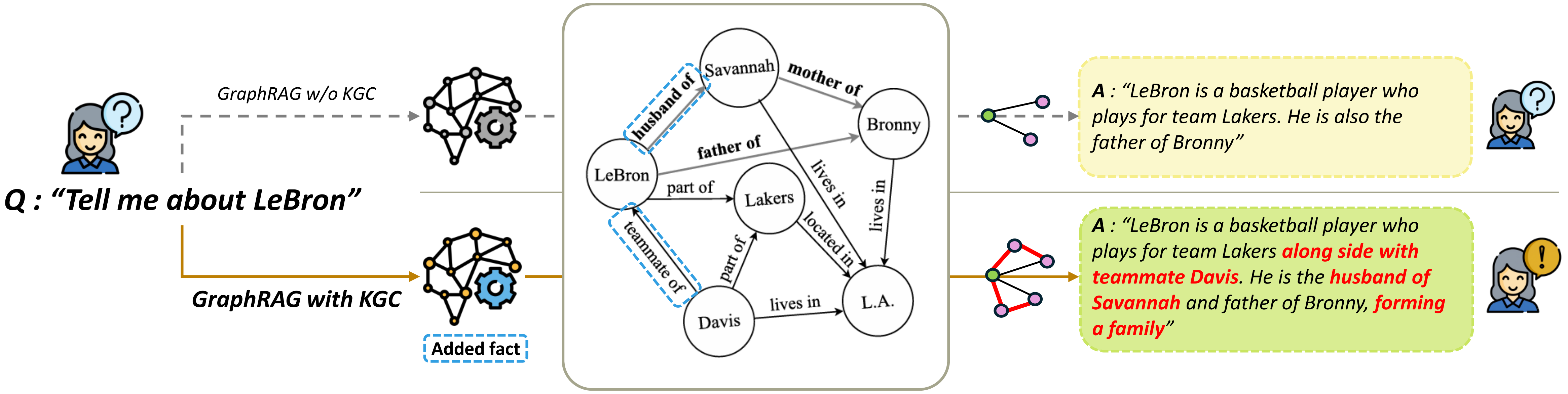


KNOWLEDGE GRAPH COMPLETION

■ Knowledge Graph Completion (KGC)

- ☐ To automatically fill missing links for inherently sparse KGs
- ☐ *Link prediction : (h, r, ?) or (?, r, t) → predict "?"*



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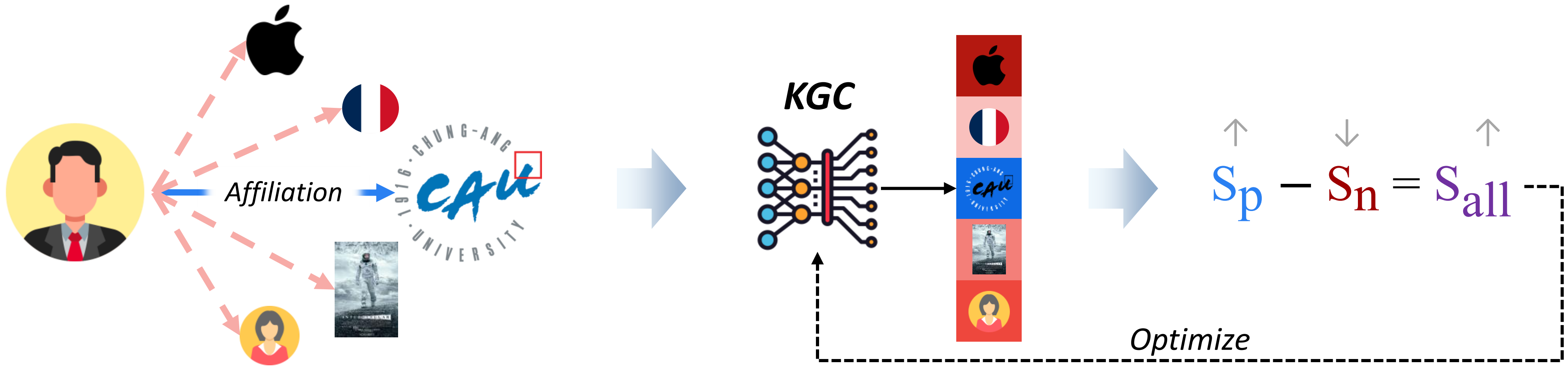
KNOWLEDGE GRAPH COMPLETION

■ Popular Strategy for KGC: Learn to Discriminate

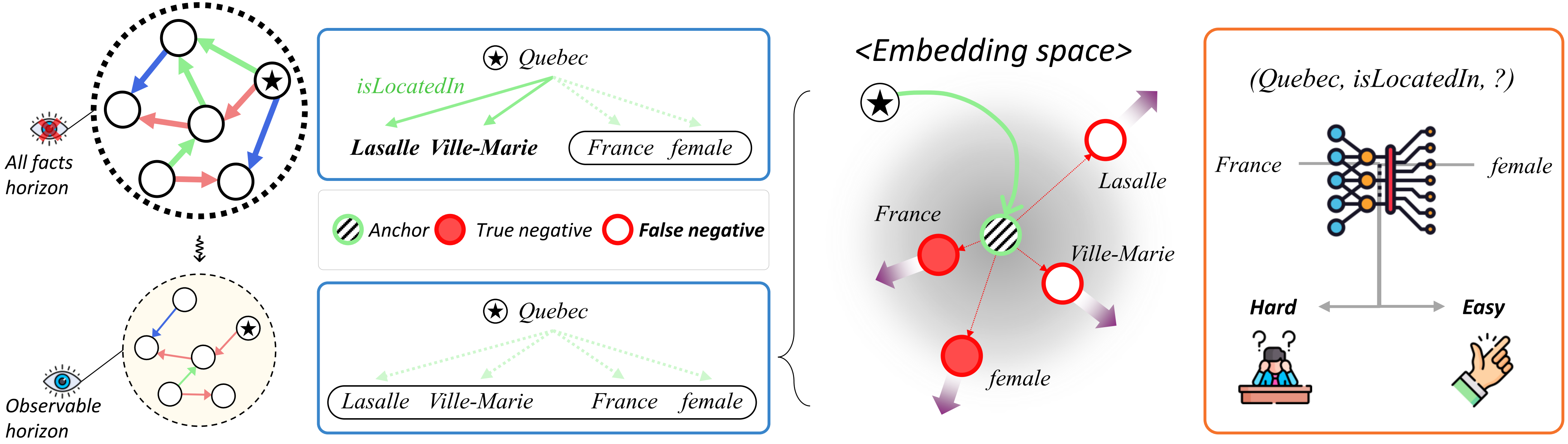
	TransE [4]				DistMult [5]				ComplEx [6]				RotatE [7]			
Metrics	MR	MRR	Hits@1	Hits@3	MR	MRR	Hits@1	Hits@3	MR	MRR	Hits@1	Hits@3	MR	MRR	Hits@1	Hits@3
w/o NS	1576.16	0.104	0.069	0.109	824.20	0.145	0.104	0.147	799.71	0.166	0.120	0.172	4176.94	0.014	0.007	0.012
w/ NS	244.71	0.291	0.200	0.323	203.40	0.291	0.206	0.318	201.00	0.316	0.223	0.347	221.07	0.310	0.218	0.343
Gain	+84%	+180%	+190%	+197%	+75%	+101%	+99%	+116%	+75%	+90%	+85%	+101%	+95%	+2,070%	+2,919%	+2,716%



S. Moon and Y. Ko, "Investigating the Effects of Negative Sampling in Knowledge Graph Completion," 2025 16th International Conference on Information and Communication Technology Convergence (ICTC)

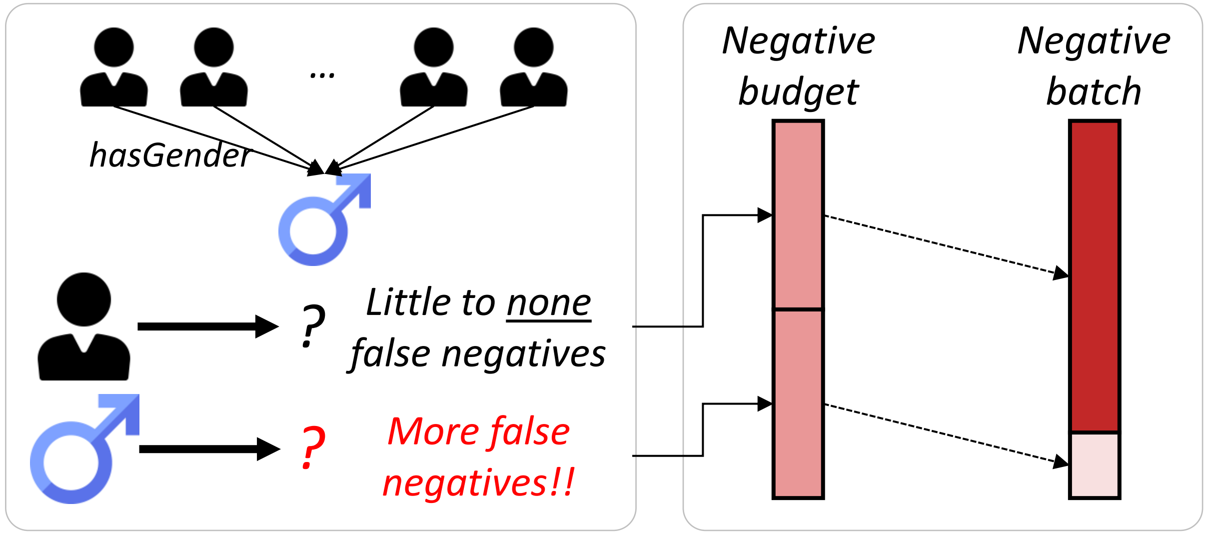
CHALLENGES IN NEGATIVE SAMPLING

■ Challenges: False Negative & Hard Negative



CHALLENGES IN NEGATIVE SAMPLING

■ (C1) Related Work: ERDNS (ISWC'23)

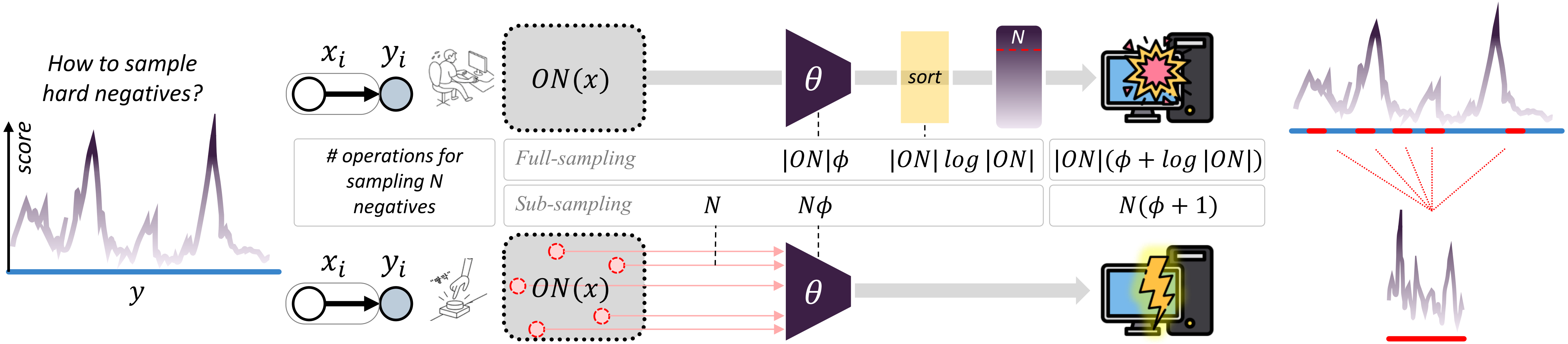


Dataset	KGE model	DistMult				ComplEx			
	Method	MRR	Hits@1	Hits@3	Hits@10	MRR	Hits@1	Hits@3	Hits@10
FB15K237	KBGAN	26.7	-	-	45.3	28.2	-	-	45.4
	NSCaching	28.3	-	-	45.6	30.2	-	-	48.1
	unif	21.9	13.1	23.4	41.1	22.7	13.7	24.5	42.1
	Bernoulli	25.9	17.4	28.3	43.3	31.0	21.2	34.3	51.3
	Self-Adv	30.9	22.1	33.6	48.4	32.2	23.0	35.1	51.0
	Subsampling	29.9	21.2	32.7	47.5	32.8	23.6	36.1	51.2
	ERDNS	**35.0**	**25.7**	**38.3**	**53.7**	**35.7**	**26.2**	**39.3**	**54.7**

Yao, Naimeng, et al. "Entity-relation distribution-aware negative sampling for knowledge graph embedding." International Semantic Web Conference. Cham: Springer Nature Switzerland, 2023.

CHALLENGES IN NEGATIVE SAMPLING

■ (C2) Related Work: Self-Adv (ICLR'19)



$$\boxed{w_{x,y}} = \frac{\exp(\alpha \cdot \theta(x,y))}{\sum_{\boxed{y' \in \mathcal{P}^-_{x,y}}} \exp(\alpha \cdot \theta(x,y'))}$$

Relative weight to high-score entities, within batch!

$$\mathcal{L}^{(x,y)} = -\log \sigma(\theta(x,y)) - \sum_{\boxed{y' \in \mathcal{P}^-_{x,y}}} \boxed{w_{x,y'}} \log(-\sigma(\theta(x,y')))$$

Similar effect as sampling from entire distribution!

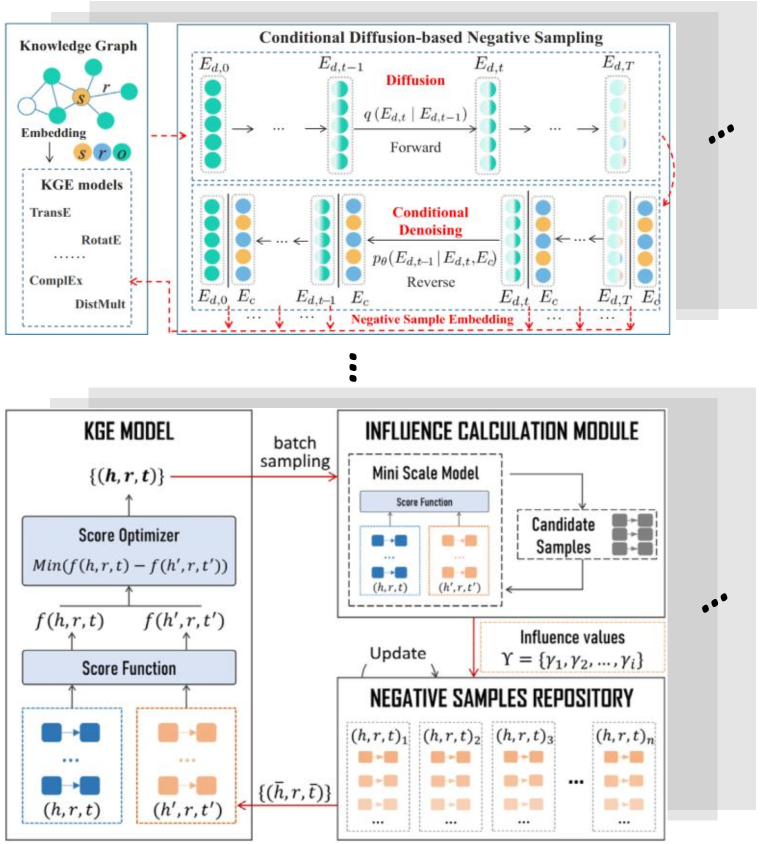
	FB15k-237		WN18RR		WN18	
	MRR	H@10	MRR	H@10	MRR	H@10
uniform	.242	.422	.186	.459	.433	.915
KBGAN (Cai & Wang, 2017)	.278	.453	.210	.479	.705	**.949**
self-adversarial	**.298**	**.475**	**.223**	**.510**	**.736**	.947

Sun, Zhiqing, et al. "Rotate: Knowledge graph embedding by relational rotation in complex space." International Conference of Learning Representations (2019).

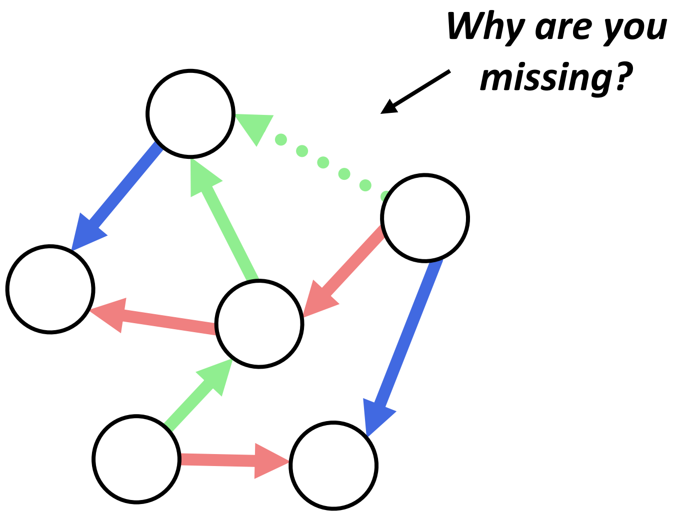
CHALLENGES IN NEGATIVE SAMPLING

■ From a More Fundamental Question

Recent Approaches



Our Approaches



BRIDGING TWO WORLDS

■ Bridge Between Closed-World KG & Open-World KG

Definition. *CWKG and OWKG*

$$\mathcal{G}^* = \bigcup_{\mathcal{G} \in W} \mathcal{G}$$

Property 1. *Surjection*

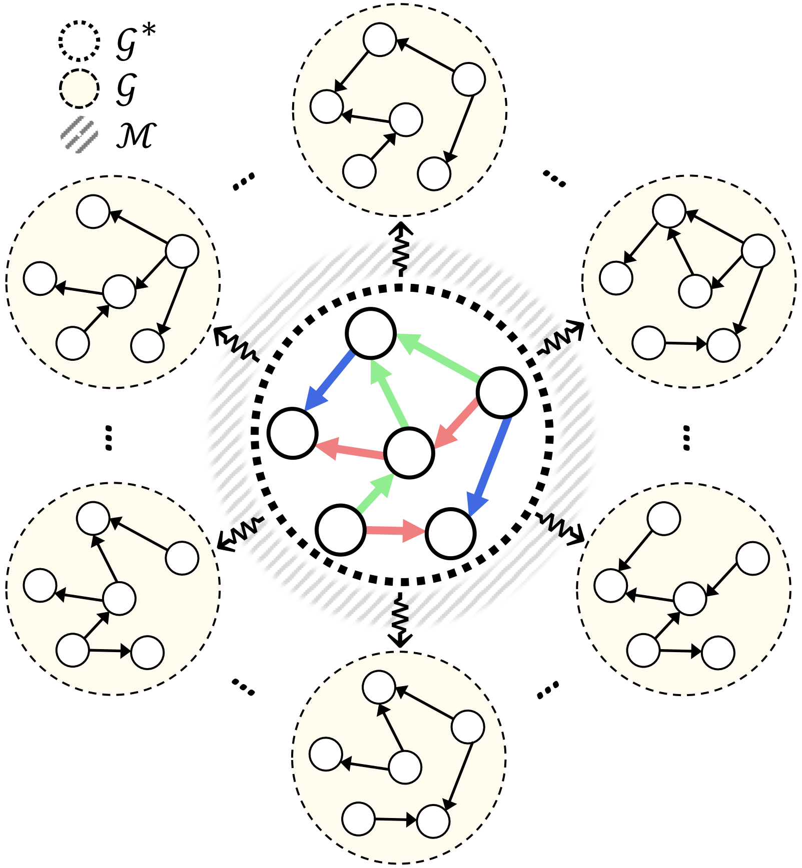
$$W \twoheadrightarrow \mathcal{G}^*$$

Property 2. *Subgraph relationship*

$$\mathcal{G} \subseteq \mathcal{G}^*$$

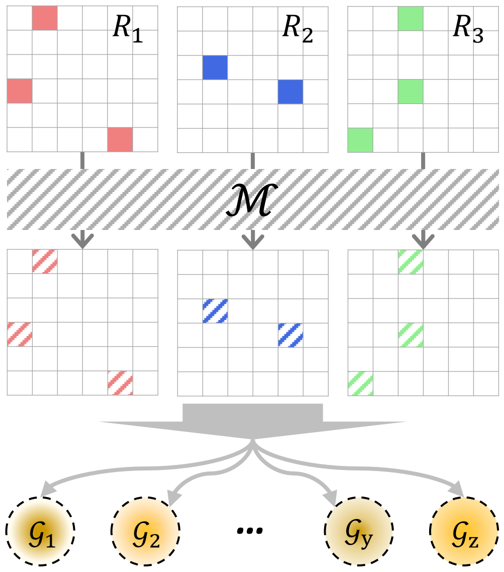
Property 3. *Fact existence*

$$f \notin \mathcal{G}^* \Rightarrow \forall \mathcal{G} \in W, f \notin \mathcal{G}$$



Proposition 1. *Observability*

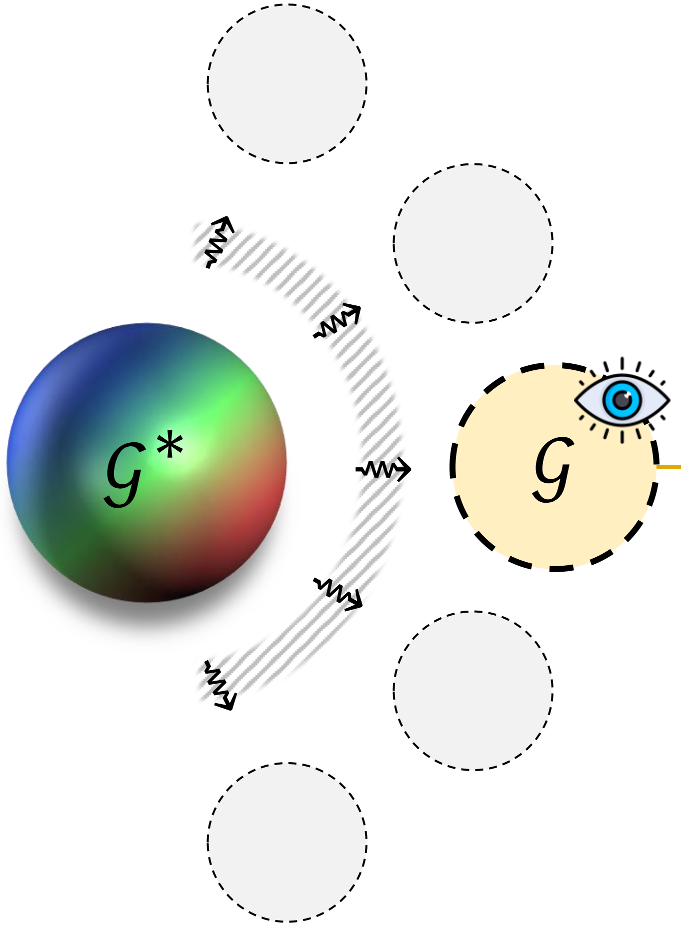
$$\forall \mathcal{G} \in W, \exists \mathcal{M}$$
$$\text{s.t.}\, \mathcal{G} \sim \mathcal{M} \odot \mathcal{G}^*$$



For simplicity, we treat $\mathcal{G}$ and its variants as both sets of facts and adjacency tensors.

BRIDGING TWO WORLDS

Probabilistic False Negative Detection



Anchor degree $d(x) = \sum_{(x,y')\in\mathcal{G}^*} \epsilon_{(x,y')} = d^*(x) \cdot \mathbb{E}_{y'\sim\mathcal{G}^*(x)}[\epsilon_{(x,y')}]$

We only need $\mathcal{M}$!!!

Entity degree $d(y) = \sum_{(x',y)\in\mathcal{G}^*} \epsilon_{(x',y)} = d^*(y) \cdot \mathbb{E}_{x'\sim\mathcal{G}^*(y)}[\epsilon_{(x',y)}]$

Missing data mechanisms for estimating $\mathcal{M}$

Why missing?

MNAR
Missing Not at Random

MAR
Missing at Random

MCAR
Missing Completely at Random

IDK

For simplicity, we treat $\mathcal{G}$ and its variants as both sets of facts and adjacency tensors.

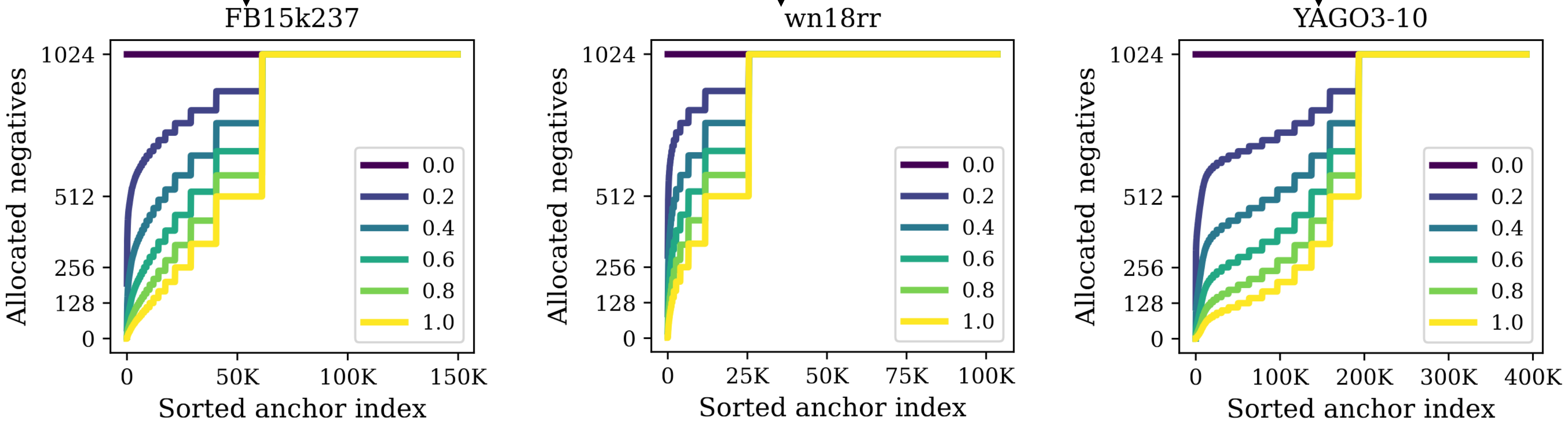
BRIDGING TWO WORLDS

■ Dynamic Negative Sampling Size

PROPOSITION 3.1. *Under uniform observable probability ϵ, the number of false negatives for equal number of draws satisfies the inequality,*

$$d(x_i) < d(x_j) \rightarrow N_{fn}(x_i) < N_{fn}(x_j).$$

$$\boxed{N(x) = \left\lfloor N_{max} \cdot \left(\frac{d_{min}}{d(x)}\right)^s \right\rceil}$$



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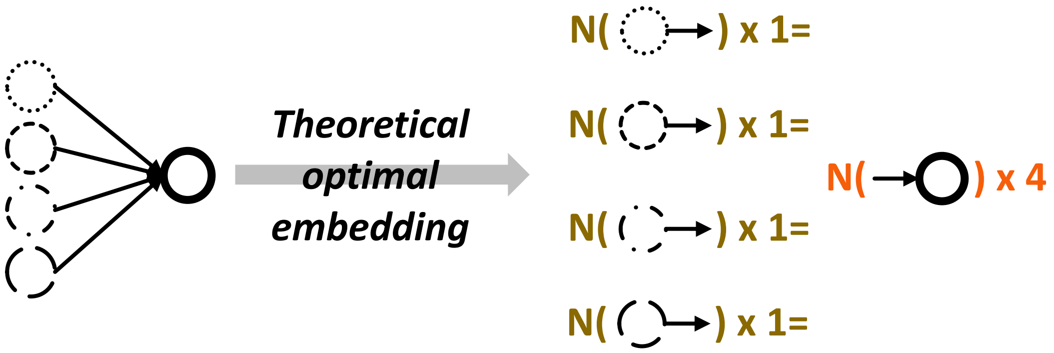
BRIDGING TWO WORLDS

■ Optimal Model Analysis (Yang et al., 2020; Kamigaito et al., 2022; Yao et al., 2023)



PROPOSITION 3.2. *To obtain optimal solution, for any anchor pair where $x_{h(i)} \in S_{x_t}$, $x_{h(i)} = (h(i), r)$ and $x_t = (t, r)$, the number of negative samples $N(x_{h(i)})$ and $N(x_t)$ should satisfy:*

$$\underline{N(x_{h(i)})d(x_{h(i)})} = \underline{N(x_t)d(x_t)}.$$

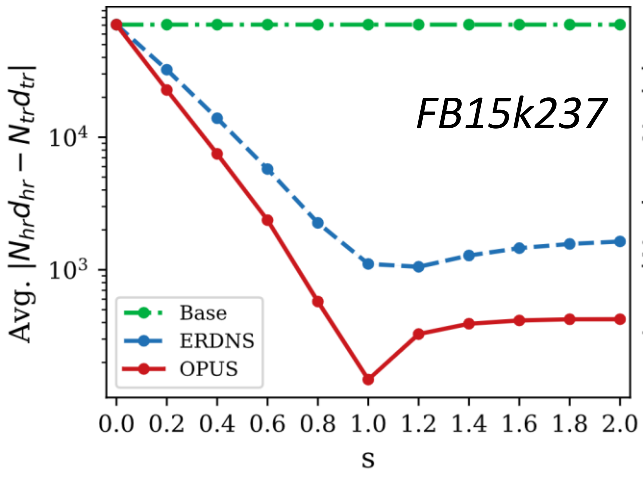
$$\boxed{diff = \Delta}$$

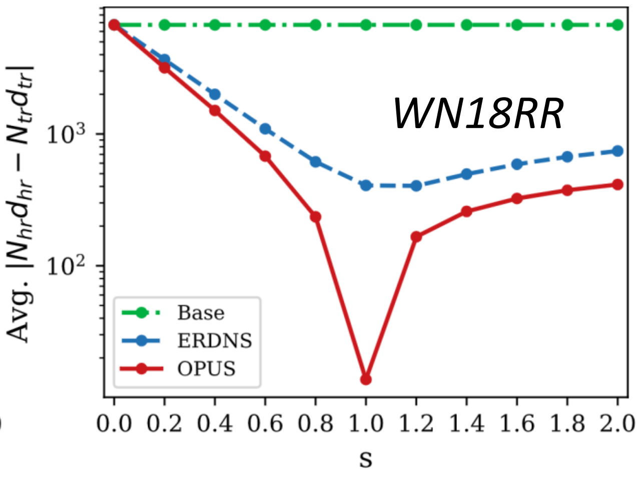
Similar conclusion can be reached for the other direction S_{x_h}.

can't reduce!

$$\boxed{\min \Delta^{(\tau)}_{ERDNS}} = |\underline{B \cdot N_{max}\left(\frac{d(x_h)^{1-\tau}}{Z_h^{(\tau)}} - \frac{d(x_h)^{1-\tau}}{Z_t^{(\tau)}}\right)} + \underline{(\gamma_h d(x_h) - \gamma_t d(x_t))}|$$

$$\boxed{\min \Delta^{(s)}_{OPUS}} = \underline{|\gamma_h d(x_h) - \gamma_t d(x_t)|} \leq \frac{1}{2} \max(d(x_h), d(x_t))$$

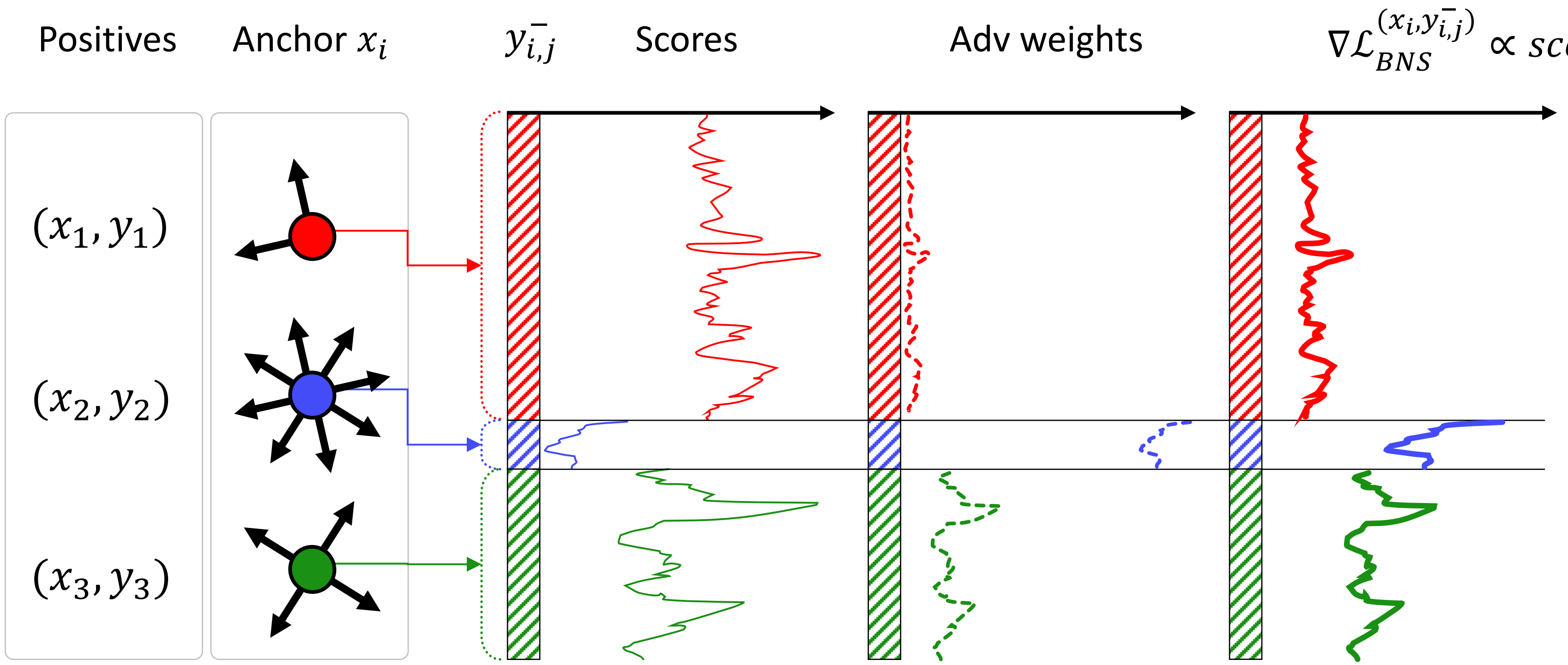




Yang, Zhen, et al. "Understanding negative sampling in graph representation learning." Proceedings of the 26th ACM SIGKDD international conference on knowledge discovery & data mining. 2020.
Kamigaito, Hidetaka, and Katsuhiko Hayashi. "Comprehensive analysis of negative sampling in knowledge graph representation learning." International conference on machine learning. PMLR, 2022.
Yao, Naimeng, et al. "Entity-relation distribution-aware negative sampling for knowledge graph embedding." International Semantic Web Conference. Cham: Springer Nature Switzerland, 2023.

SAMPLING HARD NEGATIVES

■ Problem of Dynamic NS + Self-Adversarial Weighting



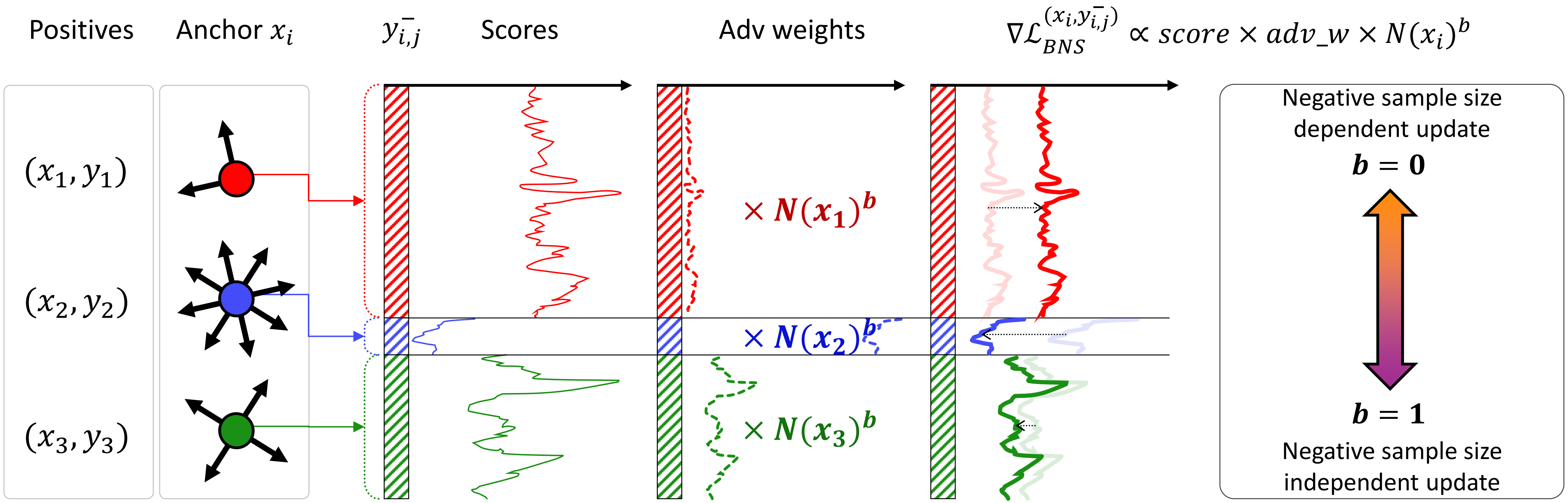
The problem is obvious

Negative size tempers loss scale

Unstable update!!!

SAMPLING HARD NEGATIVES

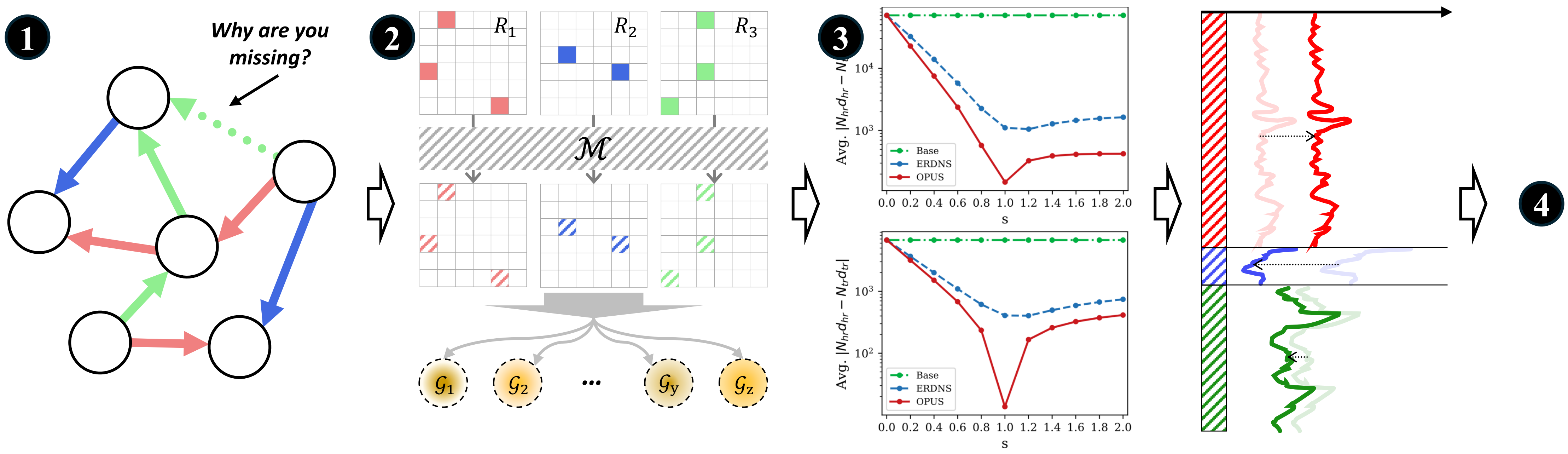
■ Re-distributing Weights to Stabilize the Posterior



SAMPLING HARD NEGATIVES

■ OPen world aware Universal Negative Sampler

- ☐ ① Raise a **fundamental question** on the existence of false negative samples
- ☐ ② Provide **theoretical background** for mitigating false negatives
- ☐ ③ Prove superiority of OPUS w.r.t **optimal embedding**
- ☐ ④ Requires **no additional** computational overhead, **faster** training time, **less** VRAM, **better** performance



EXPERIMENTS

■ Experiment Background & Research Questions

Research questions

(RQ1) Does OPUS improve performance?
(RQ2) Is OPUS efficient w.r.t time & memory?
(RQ3) Is dynamic NS effective?
(RQ4) Does OPUS mitigate false negative sampling?
(RQ5) Is OPUS robust to hyperparameters?

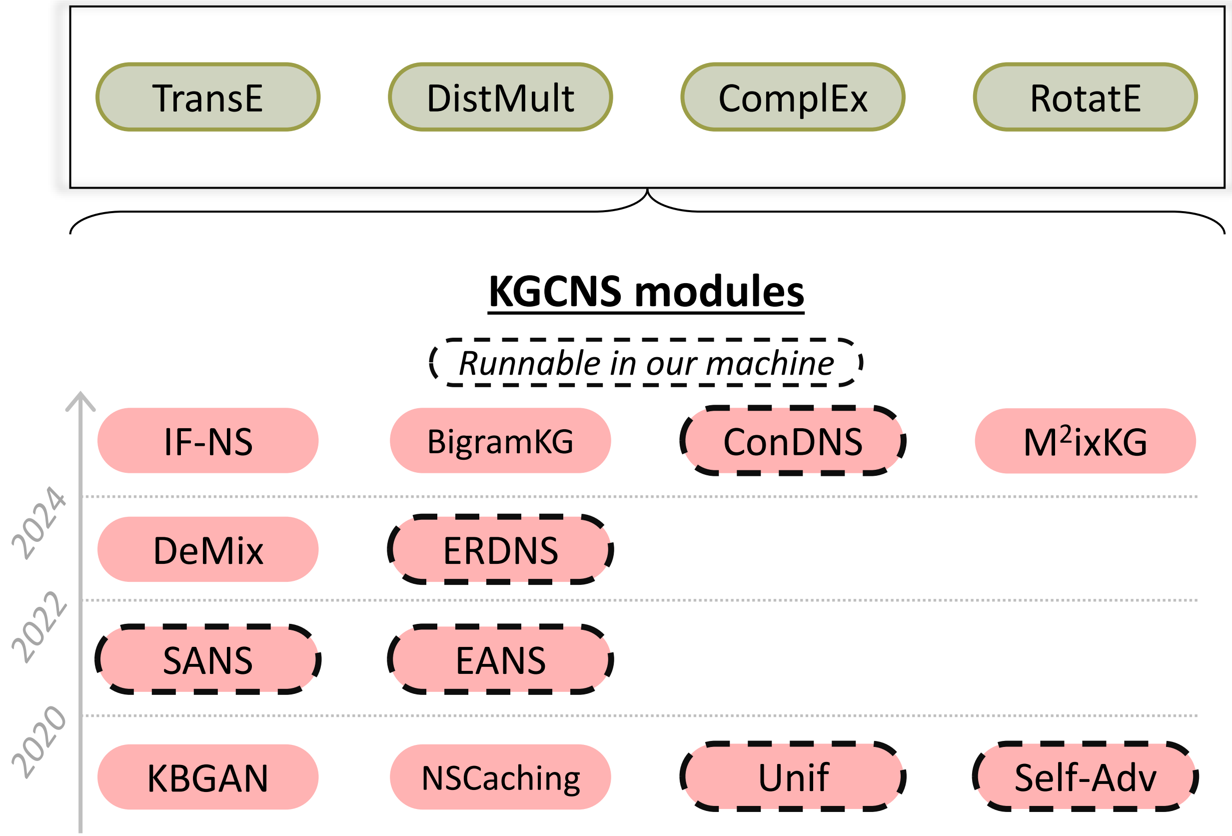
Dataset & statistics

	$\lvert\mathcal{E}\rvert$	$\lvert\mathcal{R}\rvert$	$\lvert\mathcal{T}\rvert$	$\delta(q)_{avg}$	$\delta(q)_{max}$
FB15k237	14,541	237	272,115	37.5	7,614
WN18RR	40,943	11	86,835	4.3	482

Evaluation

$$\mathrm{MRR} = \mathbb{E}[1/\mathrm{rank}] \qquad \mathrm{Hits@K} = \mathbb{E}[\mathbb{I}[\mathrm{rank} \leq \mathrm{K}]]$$

Main models



EXPERIMENTS

■ (RQ1) Performance of OPUS on FB15k237

Dataset	Method	RotatE				TransE				ComplEx				DistMult			
		MRR	Hits@1	Hits@3	Hits@10	MRR	Hits@1	Hits@3	Hits@10	MRR	Hits@1	Hits@3	Hits@10	MRR	Hits@1	Hits@3	Hits@10
FB15K237	KBGAN	-	-	-	-	29.4	-	-	49.7	28.2	-	-	45.4	26.7	-	-	45.3
	NSCaching	30.9	22.1	34.1	48.6	30.0	-	-	47.4	30.2	-	-	28.1	28.3	-	-	45.6
	DeMix-Adv	32.9	23.5	36.6	51.8	31.8	-	-	51.0	30.7	-	-	47.9	30.1	-	-	47.0
	M^2ixKG	33.7	-	-	53.2	33.1	-	-	52.9	32.3	-	-	51.6	31.7	-	-	50.3
	IF-NS	-	-	-	-	31.0	-	-	48.9	-	-	-	-	-	-	-	-
	BigramKG	34.0	-	-	53.2	33.5	-	-	53.1	32.3	-	-	51.6	31.6	-	-	49.7
	Uniform†	29.5	20.3	32.7	47.9	29.4	20.0	32.7	48.1	27.2	19.2	29.6	43.2	25.5	17.9	27.6	40.3
	Uniform SANS†	29.8	20.6	33.0	48.0	29.6	20.2	33.0	48.3	-	-	-	-	25.8	18.1	28.0	41.0
	Uniform RW SANS†	29.7	20.4	33.0	48.4	29.7	20.3	33.1	48.6	-	-	-	-	26.1	18.4	28.4	41.4
	Self-Adv†	33.6	24.0	37.3	53.1	33.0	23.1	36.9	52.7	32.2	22.9	35.4	51.2	30.8	22.0	33.6	48.4
	Self-Adv SANS†	33.6	23.9	37.3	53.0	32.6	23.0	36.5	52.1	-	-	-	-	31.1	22.4	33.9	48.7
	Self-Adv RW-SANS†	31.5	22.0	34.8	50.8	30.5	21.0	34.0	49.8	-	-	-	-	30.6	22.0	33.3	47.9
	ERDNS†	34.5	25.1	37.9	53.5	34.1	24.6	37.8	53.3	**35.7**	**26.2**	**39.2**	**54.7**	**34.9**	**25.6**	**38.3**	**53.8**
	EANS†	34.5	24.9	38.1	53.5	34.1	24.3	38.1	53.4	29.8	21.2	32.8	47.1	30.9	21.8	34.0	49.6
	ConDNS†	33.6	24.0	37.4	53.2	33.0	23.1	36.9	52.7	32.4	23.2	35.4	51.1	30.8	22.0	33.6	48.5
	OPUS†	**35.1**	**25.8**	**38.5**	**53.9**	**34.7**	**25.2**	**38.3**	**53.7**	35.0	25.7	38.3	53.9	34.5	25.3	37.9	53.0

EXPERIMENTS

■ (RQ1) Performance of OPUS on WN18RR

Dataset	Method	RotatE				TransE				ComplEx				DistMult			
		MRR	Hits@1	Hits@3	Hits@10	MRR	Hits@1	Hits@3	Hits@10	MRR	Hits@1	Hits@3	Hits@10	MRR	Hits@1	Hits@3	Hits@10
WN18RR	KBGAN	-	-	-	-	18.6	-	-	45.4	42.9	-	-	47.0	38.5	-	-	44.3
	NSCaching	47.1	42.9	48.9	55.3	20.0	-	-	47.8	44.6	-	-	50.9	41.3	-	-	45.5
	DeMix-Adv	47.9	42.8	49.2	57.6	22.0	-	-	52.1	46.8	-	-	55.2	43.9	-	-	53.5
	M^2ixKG	47.7	-	-	57.5	23.0	-	-	54.1	47.4	-	-	56.2	44.6	-	-	54.1
	IF-NS	-	-	-	-	23.6	-	-	50.9	-	-	-	-	-	-	-	-
	BigramKG	47.7	-	-	57.3	22.3	-	-	52.9	47.3	-	-	56.8	44.3	-	-	54.3
	Uniform†	47.5	43.3	49.0	55.7	20.5	1.8	35.3	50.0	46.5	42.6	48.4	53.8	42.8	37.7	44.6	52.8
	Uniform SANS†	47.8	43.6	49.3	56.0	20.4	1.8	35.1	50.0	-	-	-	-	40.9	37.7	42.3	46.7
	Uniform RW-SANS†	48.1	43.6	49.5	56.9	22.0	3.1	37.1	51.1	-	-	-	-	41.7	37.3	43.7	49.7
	Self-Adv†	47.7	43.4	49.3	56.3	22.5	1.4	40.2	53.1	46.9	42.6	48.8	55.4	43.9	39.3	45.4	53.3
	Self-Adv SANS†	47.7	43.5	49.3	56.0	22.5	1.5	40.2	53.2	-	-	-	-	37.8	35.3	38.8	42.3
	Self-Adv RW-SANS†	48.3	43.7	49.9	57.2	22.6	3.4	38.3	51.5	-	-	-	-	39.9	37.1	41.3	45.0
	ERDNS†	47.7	43.2	49.4	56.7	23.9	3.0	41.7	53.2	48.3	44.6	49.9	55.7	45.5	41.0	46.8	54.8
	EANS†	48.8	44.4	50.3	57.8	22.9	2.6	39.9	53.6	45.4	41.4	47.4	53.4	43.3	39.2	44.6	52.1
	ConDNS†	47.7	43.1	49.4	56.7	22.6	1.6	40.4	53.2	46.9	42.6	48.9	55.9	43.8	39.2	45.5	53.2
	OPUS†	**49.1**	**44.6**	**50.7**	**58.1**	**25.5**	**5.5**	**42.5**	**55.2**	**49.5**	**45.4**	**50.9**	**57.3**	**45.8**	**41.4**	**47.3**	**54.9**

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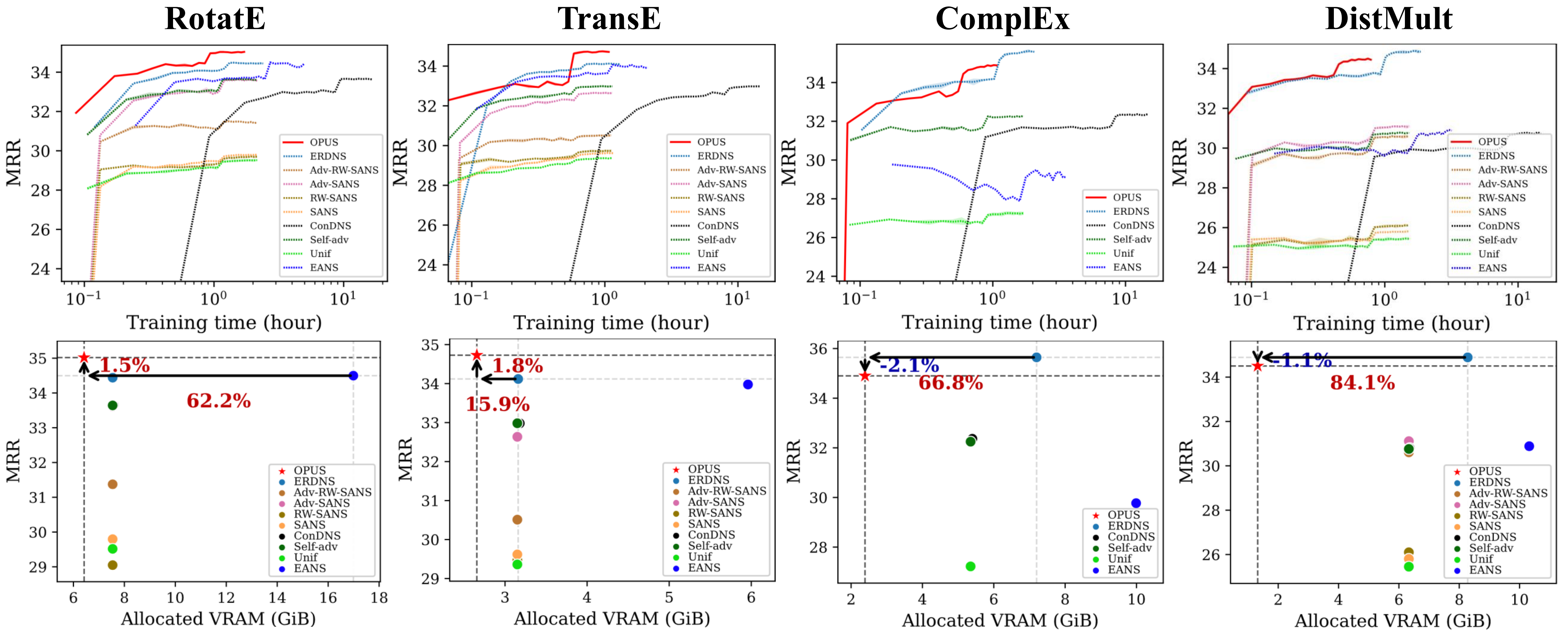
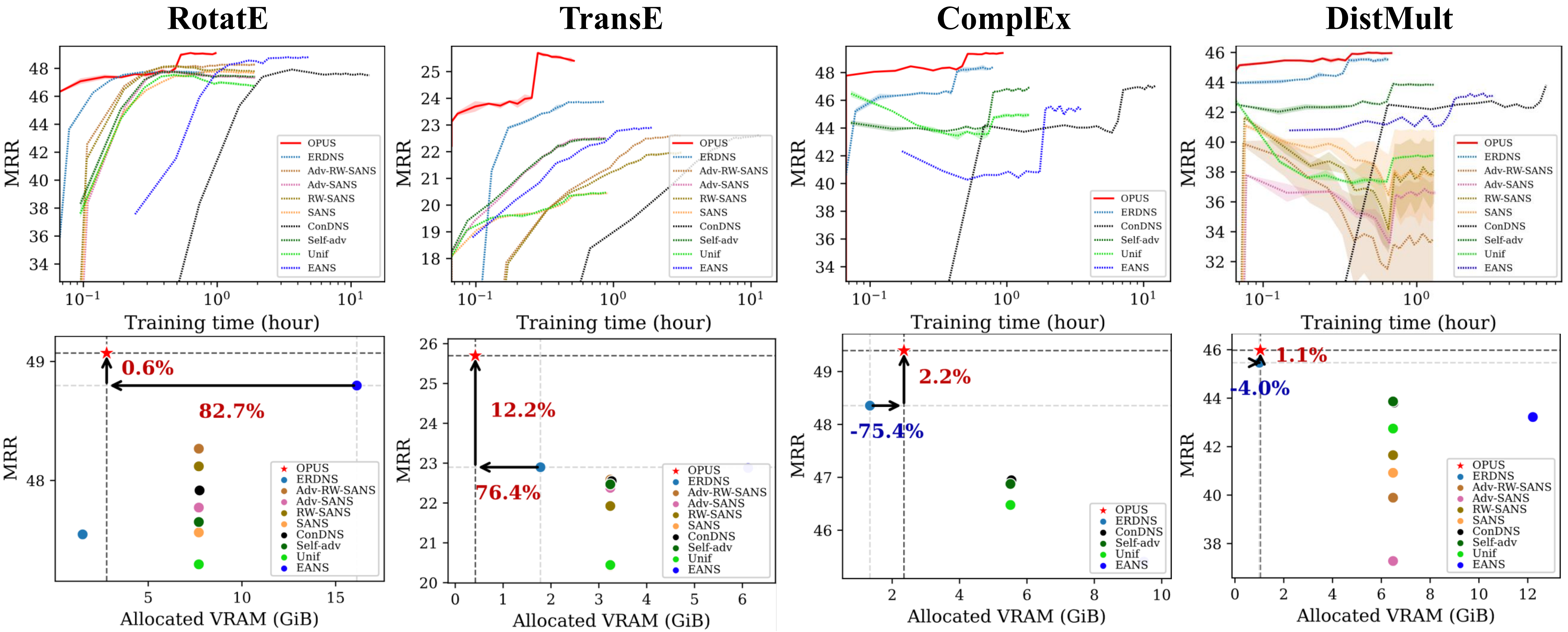
EXPERIMENTS

■ (RQ2) Efficiency of OPUS on FB15k237



EXPERIMENTS

(RQ2) Efficiency of OPUS on WN18RR



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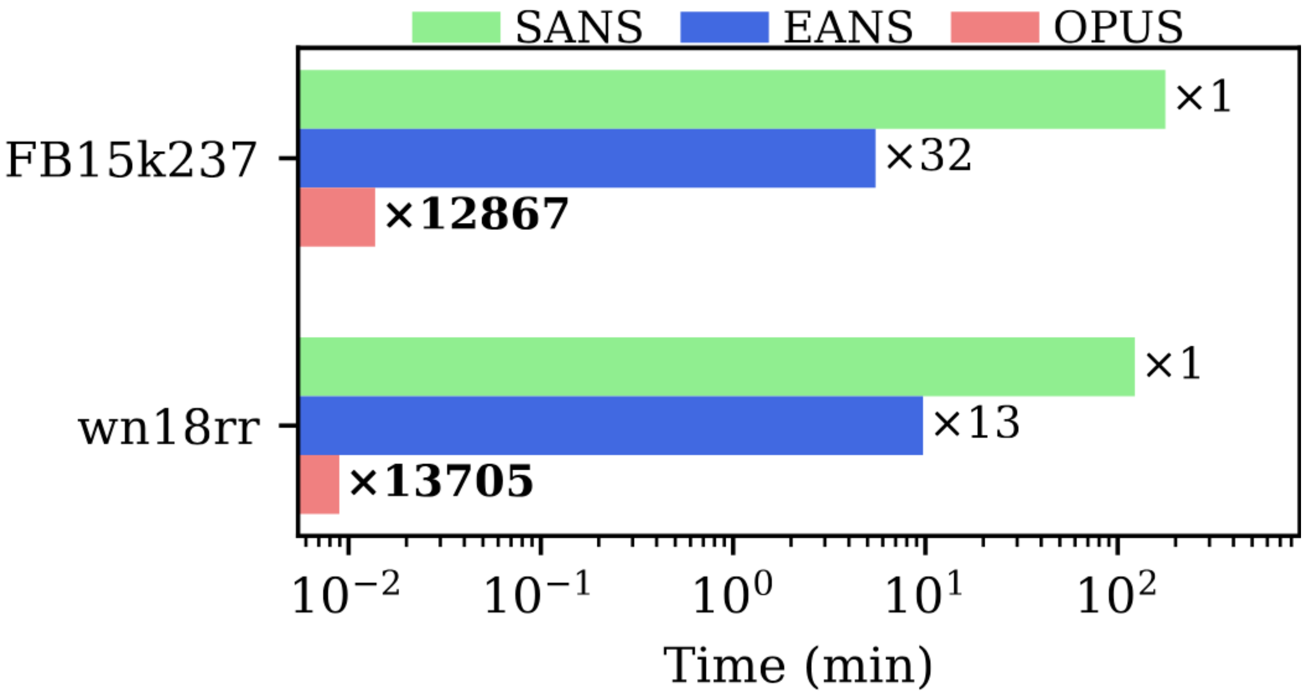
EXPERIMENTS

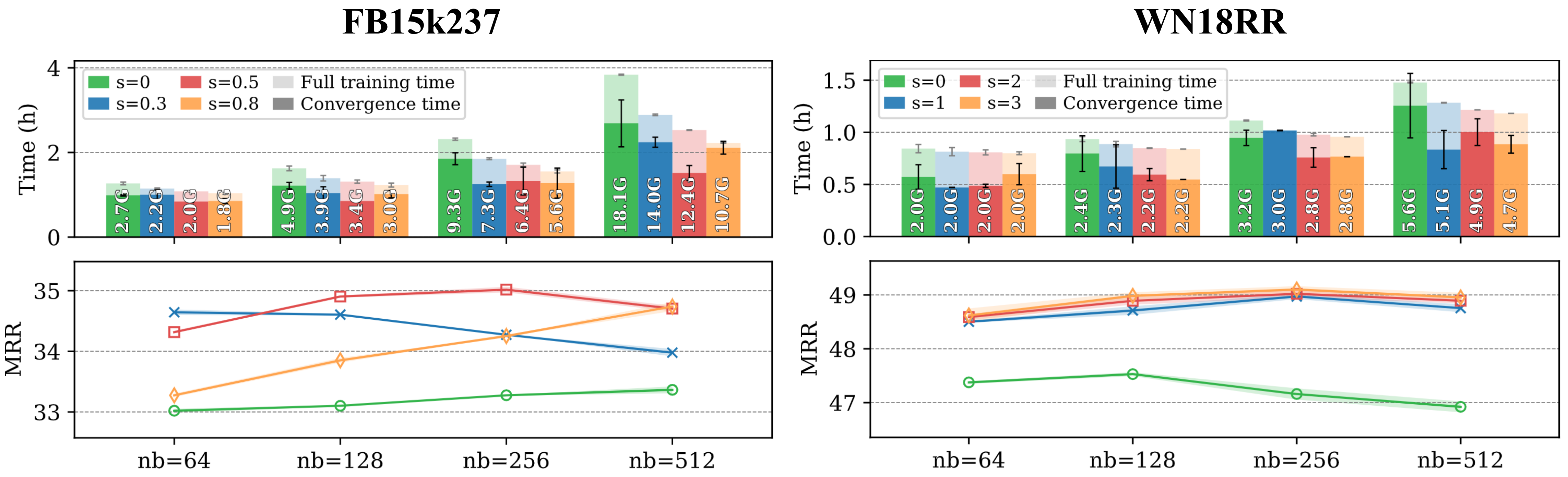
■ (RQ2) Required Data Processing Time

- ☐ SANS (*before training*) : Random walk
- ☐ EANS (*during training*) : KNN
- ☐ OPUS (*before training*) : Negative sample size allocation



EXPERIMENTS

■ (RQ3) Effect of Dynamic Negative Sampling



*Base model is RotatE

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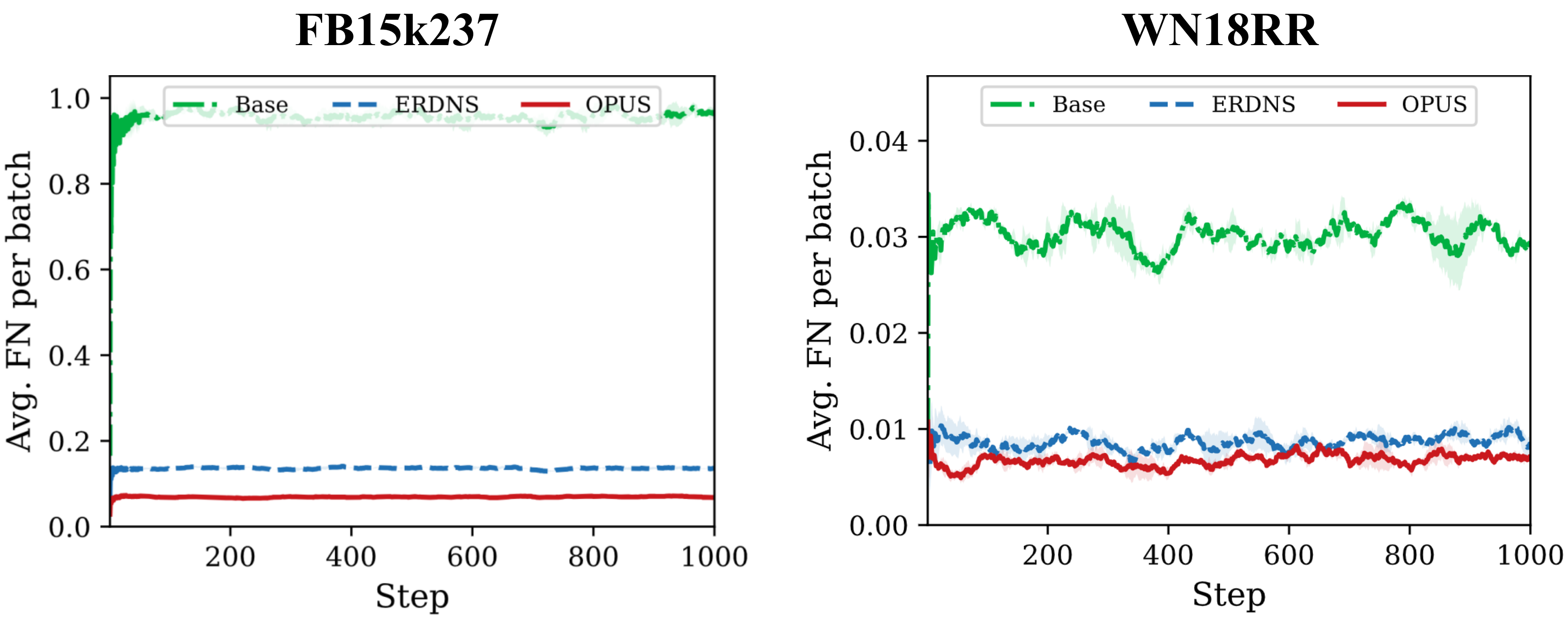
EXPERIMENTS

■ (RQ4) Mitigating False Negative Sampling

	Base	ERDNS	OPUS
PB	1024	1024	1024
NB	256	256	256
S	0.0	0.5	0.5



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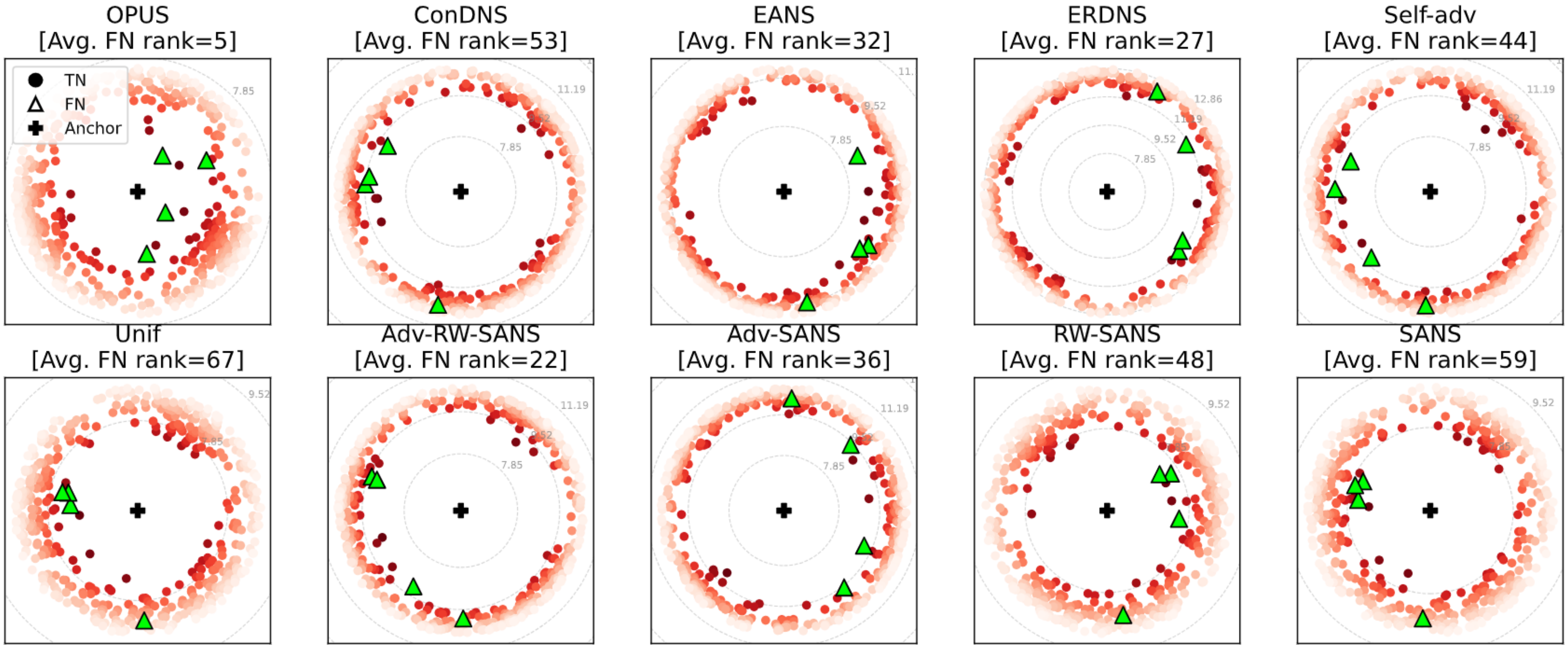
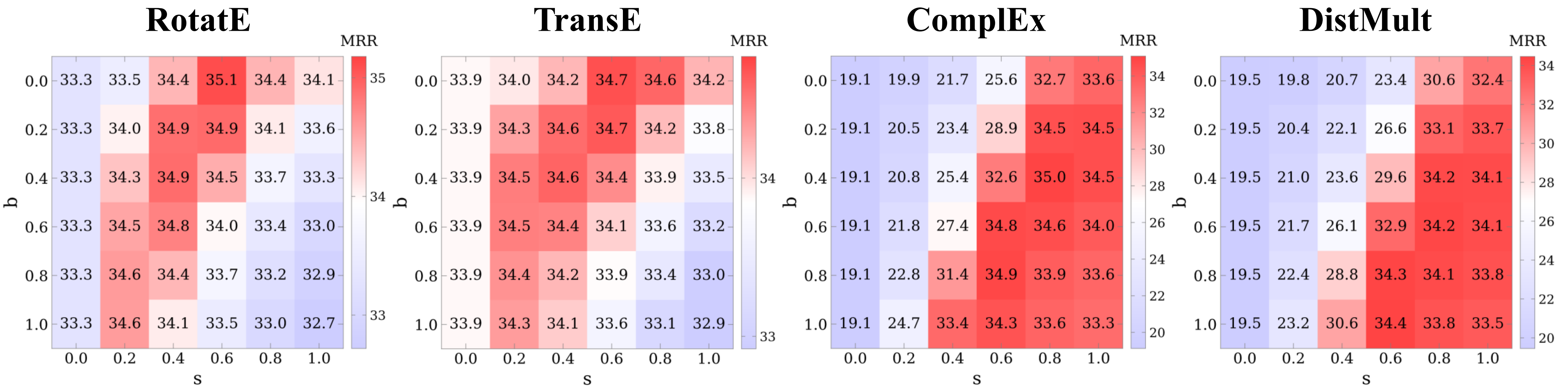
EXPERIMENTS

■ (RQ4) PCA Case Study with TN & FN Positions



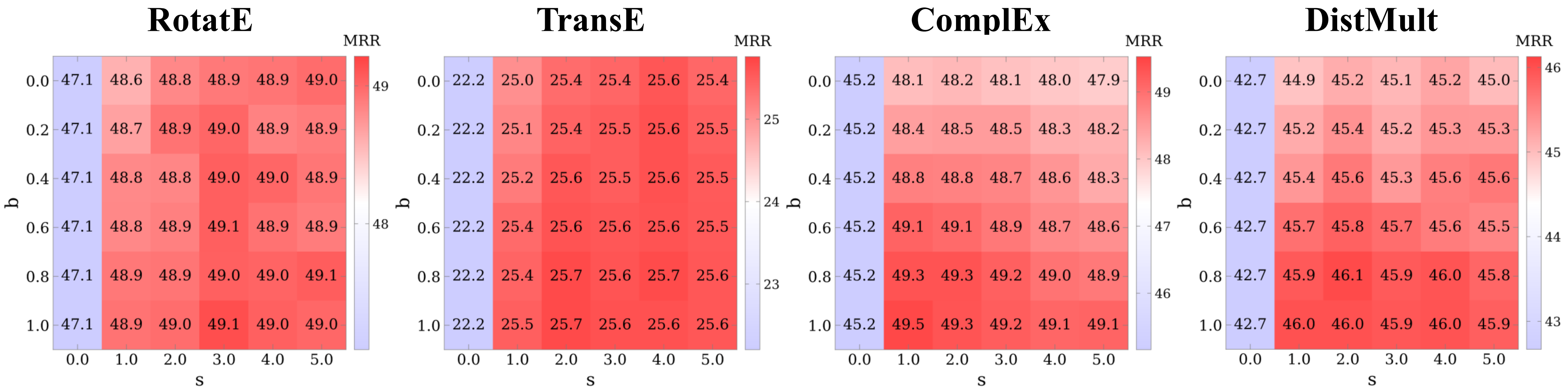
EXPERIMENTS

■ (RQ5) Robustness to Hyperparameter on FB15k237



EXPERIMENTS

■ (RQ5) Robustness to Hyperparameter on WN18RR



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CONCLUSION

■ Negative Sampling for KGC

- ☐ Incredibly simple, yet highly effective

■ Fundamental Question

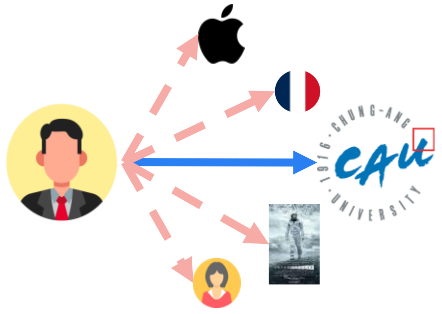
- ☐ *“Why are some links missing?”*
- ☐ Probabilistically mitigate false negative sampling
- ☐ Simple method, easy to extend

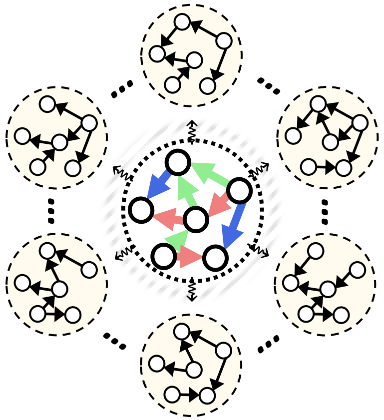
■ Theoretical Analysis

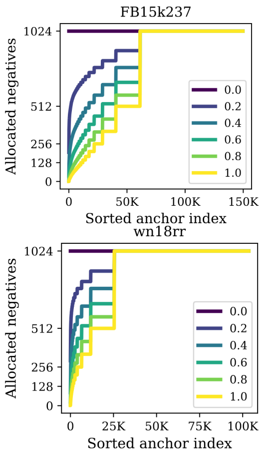
- ☐ OPUS under optimal embedding

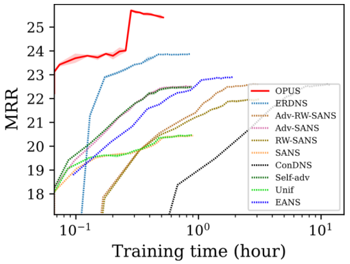
■ Experimental Results

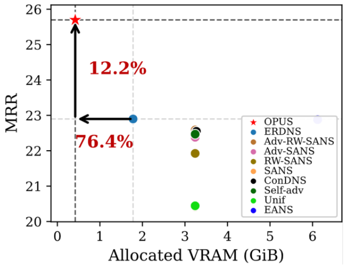
- ☐ Stands out compared to baselines
- ☐ Highly efficient in terms of time & memory
- ☐ Successfully mitigate false negatives











Q&A